

Practical exercises

— ordering and multiplying decimals

A Cambridge insert: putting decimals in order, and multiplying them without a calculator.

LESSON 11–12

2 × 40 MIN

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S8 4.1–4.2

LESSON CLOCK

15'

25'

25'

20'

5'

Ordering	15 min
Multiplying	25 min
Dividing	25 min
Checking	20 min
Homework	5 min

LEARNING OBJECTIVES

- Order decimals by comparing place value, digit by digit.
- Multiply two decimals and place the point by counting decimal places.
- Divide a decimal by a decimal by making the divisor whole.
- Check every answer by estimation.

TEXTBOOK

National	pp. 32–35
Cambridge	Stage 8, pp. 36–44
Workbook	Exercise 4.1–4.2

01

Explanation

Ordering

Compare decimals digit by digit from the left, not by counting digits.

PAIR	LARGER	WHY
0.7 and 0.65	0.7	7 > 6 in the tenths
3.09 and 3.1	3.1	1 > 0 in the tenths
0.4 and 0.400	equal	trailing zeros change nothing

MORE DIGITS DOES NOT MEAN LARGER

0.65 has more digits than 0.7 and is smaller. Writing both with the same number of decimal places — 0.70 and 0.65 — removes the illusion entirely.

Multiplying

STEP	WHAT TO DO
1	ignore the points and multiply as whole numbers
2	count the decimal places in both factors
3	put that many decimal places in the answer

Example. 0.34×0.6 . Whole numbers: $34 \times 6 = 204$. Places: $2 + 1 = 3$. Answer: 0.204 .

PRODUCT	WHOLE NUMBERS	PLACES	ANSWER
0.3×0.4	12	2	0.12
1.2×0.05	60	3	0.060
2.5×0.4	100	2	1.00

MULTIPLYING TWO DECIMALS BELOW 1 GIVES A SMALLER ANSWER

$0.3 \times 0.4 = 0.12$ — smaller than both. It is the same surprise as division by 0.1 , seen from the other side, and the estimate catches it every time.

Dividing

Make the **divisor** a whole number by multiplying both numbers by the same power of ten. The quotient is unchanged.

$$\frac{4.8}{0.06} = \frac{480}{6} = 80$$

DIVISION	REWRITTEN	ANSWER
$7.2 \div 0.9$	$72 \div 9$	8
$0.35 \div 0.7$	$3.5 \div 7$	0.5
$1.44 \div 0.12$	$144 \div 12$	12

BOTH NUMBERS MUST BE MULTIPLIED

Multiplying only the divisor changes the answer. Writing the division as a fraction makes it obvious: a fraction is unchanged only when top and bottom are treated alike.

Checking

CHECK	QUESTION TO ASK
size	is the answer roughly what the estimate said?
direction	should multiplying have made it bigger or smaller?
last digit	does 0.34×0.6 end in the last digit of 4×6 ?
reverse	does the answer times the divisor give the dividend back?

THE ESTIMATE IS THE CHECK

Every answer in this lesson can be verified in ten seconds by rounding to one significant figure. Doing so consistently is worth more than any amount of careful digit-copying.

02 Worked examples

EXAMPLE 1 Put in ascending order: 0.7, 0.65, 0.702, 0.6.

1

Write with three decimal places: 0.700, 0.650, 0.702, 0.600.

2

Compare as whole numbers: 600, 650, 700, 702.

3

$0.6 < 0.65 < 0.7 < 0.702$

4

Equalising the places is the whole method.

ANSWER

0.6, 0.65, 0.7, 0.702

EXAMPLE 2 Compute 0.34×0.6 .

- ① $34 \times 6 = 204$
Whole numbers.
- ② Decimal places: $2 + 1 = 3$.
- ③ 0.204
- ④ Estimate: $0.3 \times 0.6 = 0.18$ ✓

ANSWER

 0.204 **EXAMPLE 3** Compute $4.8 \div 0.06$.

- ① Multiply both by 100.
- ② $480 \div 6$
- ③ $= 80$
- ④ Check: $80 \times 0.06 = 4.8$ ✓

ANSWER

 80 **03 Interactive model**

Write 0.7 and 0.65 on the board and ask which is larger; collect the wrong answer, then write 0.70 and 0.65 and let the class correct itself.

This model could not start: den is not a function

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
To order	Tartiblash	Упорядочить
Ascending order	O'sish tartibi	По возрастанию
Descending order	Kamayish tartibi	По убыванию
Place value	Xona qiymati	Разрядное значение
Divisor	Bo'luvchi	Делитель
Dividend	Bo'linuvchi	Делимое
Trailing zero	Oxirgi nol	Конечный ноль
Product	Ko'paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which is larger, 0.7 or 0.65 ?

0.4 and 0.400 are:

0.3×0.4 equals:

0.34×0.6 has how many decimal places?

1

2

3

4

$4.8 \div 0.06$ equals:

0.8

8

80

800

In a division, you may multiply:

only the divisor

only the dividend

both by the same number

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Larger: 0.7 or 0.65

ANSWER 0.7

2. Larger: 3.09 or 3.1

ANSWER 3.1

3. 0.3×0.4

ANSWER 0.12

4. 0.2×0.5

ANSWER 0.1

5. $7.2 \div 0.9$

ANSWER 8

6. $0.35 \div 0.7$

ANSWER 0.5

7. $1.44 \div 0.12$

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. Ascending: $0.7, 0.65, 0.702, 0.6$

ANSWER $0.6, 0.65, 0.7, 0.702$

2. 0.34×0.6

ANSWER 0.204

3. 1.2×0.05

ANSWER 0.06

4. 2.5×0.4

ANSWER 1

5. $4.8 \div 0.06$

ANSWER 80

6. $0.048 \div 0.8$

ANSWER 0.06

7. 1.25×0.08

ANSWER 0.1

Hard · stretch and reason

7 PROBLEMS

1. $0.25 \times 0.4 \times 0.5$

ANSWER 0.05

2. $(0.2)^3$

ANSWER 0.008

3. $6.25 \div 0.025$

ANSWER 250

4. Descending: $0.09, 0.1, 0.089, 0.091$

ANSWER 0.1, 0.091, 0.09, 0.089

5. $0.6 \times 0.6 - 0.4 \times 0.9$

ANSWER 0

6. A cloth of 0.85 m costs $34\ 000$ so'm: the price per metre

ANSWER 40 000

7. $12.5 \times 0.008 \div 0.05$

ANSWER 2

07 Homework**Homework — 5 tasks**

Estimate every product and quotient before computing it.

- Put in ascending order: $0.45, 0.5, 0.409, 0.451$.
- Compute 0.28×0.7 and 1.5×0.06 .
- Compute $9.6 \div 0.08$ and $0.56 \div 0.7$.
- Compute $(0.3)^2$ and $(0.1)^3$.
- 1.2 kg of rice costs $18\ 000$ so'm. Find the price of 1 kg .